

Referee package – DR-2 for the lattice class: an unconditional, decoupling-free additive-energy bound on the sphere

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

This referee document concerns DR-2 (the additive-energy / discrete-restriction bound) for the lattice competitor class. It is a harmonic-analysis / additive-combinatorics result, of interest independently of TECT. **Scope:** $Q = \Lambda \cap \{|x|^2 = R\}$ a fixed crystallographic lattice shell. **Not claimed:** the arbitrary (non-lattice) Q case remains OPEN (the decoupling-based route is T4 conditional and not load-bearing); this package does not by itself close the full Reading-H C_{full} theorem. **Grade:** T7-candidate on the lattice class. The 7-script clean-run is operator-CONFIRMED (2026-06-12), and the Lemma NT pin is now IN-BUNDLE (Sec. 3: proof note `dr2-lemma-nt-inbundle-260612-v1.0` – unramified case proved in full, ramified corner cited + outlined + exhaustively machine-covered on the applied domain); the confirmed-T7 promotion awaits the operator’s review of the pin note.

2. Definitions and the result

Definitions (patch v1.1-2). Let $Q \subset \{|x|^2 = R\}$ be finite with $N = |Q|$. The additive energy and the sum-representation counts are

$$E_+(Q) = \#\{(a, b, c, d) \in Q^4 : a + b = c + d\}, \quad r_Q(m) = \#\{(a, b) \in Q^2 : a + b = m\},$$

and the maximal proper sum-circle occupancy is $T'(Q) = \max_{m \neq 0} r_Q(m)$. Lemma A (sum-circle richness) is the chain

$$E_+(Q) = \sum_m r_Q(m)^2 \leq r_Q(0)^2 + T'(Q) \sum_{m \neq 0} r_Q(m) \leq (1 + T'(Q))N^2,$$

using $r_Q(0) \leq N$ and $\sum_m r_Q(m) = N^2$.

The result. For $Q = \Lambda \cap \{|x|^2 = R\}$ (Λ a fixed crystallographic lattice, rational after clearing denominators), the additive energy obeys

$$E_+(Q) \leq (1 + C_\Lambda \max_{m \neq 0} d(4R - |m|^2))N^2 \leq (1 + C_\Lambda C_\varepsilon R^\varepsilon)N^2,$$

decoupling-free, where $d(\cdot)$ is the divisor function and C_Λ is the representation constant of Lemma NT below ($C_\Lambda \leq 6$ once the textbook lemma is imported). For Gauss-typical shells $R \sim N^2$ this gives, after redefining ε ($R^\varepsilon \sim N^{2\varepsilon}$), $E_+ \leq_\varepsilon N^{2+\varepsilon}$.

3. Structure of the argument

Circle-to-binary-form reduction (patch v1.1-3). Fix $m \neq 0$ and suppose $a + b = m$ with $|a|^2 = |b|^2 = R$. From $b = m - a$, $R = |b|^2 = |m|^2 - 2a \cdot m + |a|^2$ forces $2a \cdot m = |m|^2$. Substituting $y = 2a - m$,

$$y \cdot m = 2a \cdot m - |m|^2 = 0, \quad |y|^2 = 4|a|^2 - 2(2a \cdot m) + |m|^2 = 4R - |m|^2, \quad y \equiv -m \pmod{2},$$

and $a \mapsto y$ is injective. Hence $r_Q(m)$ is bounded by the HOMOGENEOUS representation count of $D = 4R - |m|^2$ on the rank-two lattice section $L_m = \{y \in \Lambda_{\text{int}} : y \cdot m = 0\}$ (machine check: `dr2_divcirc_proof.py`, substitution exact, injectivity verified).

Lemma NT (textbook import; patch v1.1-4). For every rank-two lattice section L_m as above, with induced positive-definite integral binary quadratic form q of content g and primitive part q_0 of discriminant $\Delta_0 < 0$,

$$r_{L_m}(D) = \#\{y \in L_m : |y|^2 = D\} \leq C_{\text{NT}} d(D), \quad C_{\text{NT}} = w(\Delta_0) \leq 6,$$

where $w(\Delta_0)$ is the number of automorphs (6 for $\Delta_0 = -3$, 4 for $\Delta_0 = -4$, 2 otherwise). Provenance: $r_q(D) = r_{q_0}(D/g)$ (zero unless $g \mid D$), and Dirichlet's representation formula gives the TOTAL count over the full form class group of discriminant Δ_0 as $w(\Delta_0) \sum_{d \mid D'} \chi_{\Delta_0}(d) \leq w(\Delta_0) d(D') \leq w(\Delta_0) d(D)$; a single class's count is at most the total. This is textbook material (Dirichlet's class-number/representation formula; e.g. Landau, or Cox, *Primes of the Form $x^2 + ny^2$*), but the package treats it as an IMPORT: the load-bearing chain is written with the constant C_Λ , and $C_\Lambda \leq 6$ holds once Lemma NT is cited or proved in the bundle. **Pin status (v1.4):** the lemma is now PINNED IN-BUNDLE by the proof note `dr2-lemma-nt-inbundle-260612-v1.0`: the unramified case (every odd $p \mid D$ with $p^2 \nmid \Delta_0$) is proved in full, elementarily, with the explicit constant $r_{L_m}(D) \leq 24 d(D)^2$ (sufficient for the chain, which needs only $d(D)^2 \ll_\varepsilon D^\varepsilon$); the ramified corner carries a book-level citation (Landau/Cassels/Cox), a conductor-descent outline, and EXHAUSTIVE applied-domain machine coverage. Machine evidence: $r \leq 6d$ holds for EVERY distinct sum m of every applied shell – 2,644,976 sums, worst ratio 0.250, 24,096 ramified instances all passing (worst 0.119) (`dr2_lemma_nt_exhaustive.py` 5/5); the richest-circle spot check (`dr2_divcirc_proof.py` 4/4) is thereby subsumed.

Assembly. Lemma A + the reduction + Lemma NT give $T'(Q) \leq C_\Lambda d(4R - |m^*|^2) \ll_\varepsilon R^\varepsilon$ (divisor bound, uniform in m), hence the stated result — no ℓ^2 -decoupling is used. The conditional decoupling route (`dr2_decoupling_iteration.py` / `dr2_decoupling_exponent.py`) is retained as a CROSS-CHECK only and is not load-bearing.

4. Numerical reproduction

Entry-script manifest (patch v1.1-1; amended v1.4: 8 scripts). Eight scripts, each self-contained with self-test asserts, all exit 0 (AI-side clean-run 2026-06-12, 38/38 asserts PASS in total; operator clean-run 2026-06-12 confirmed the first seven at 33/33; the 8th, `dr2_lemma_nt_exhaustive.py`, is NEW with the v1.4 Lemma NT pin and awaits the operator's one-script re-run, 1.2s):

1. `codes/vacuum/dr2_circle_richness.py` (5/5) — Lemma A (sum-circle richness) and Corollary 1.
2. `codes/vacuum/dr2_divcirc_proof.py` (4/4) — the substitution $y = 2a - m$ (exact, injective), $T' \leq r_{L_m}(4R - |m|^2) \leq 6d(4R - |m|^2)$, worst $T'/6d = 0.25$.
3. `codes/vacuum/dr2_lattice_divisor.py` (7/7; v2.0.0 runtime patch) — lattice-divisor scaling: T'/N falls $0.107 \rightarrow 0.024$ across shells $R = 101\text{--}9974$; $E_+/N^2 \leq 5.3$. The v1.0.0 occupancy loop timed out on $R = 9974$ in referee clean-runs; v2.0.0 replaces it by one vectorized unique-count

pass via the exact full-shell identity $\#(Q \cap C_m) = r_Q(m)$, machine-verified against the retained v1.0.0 reference on $R \leq 909$ and pinned to the archived $R = 9974$ values ($N = 2040$, $T' = 48$, $E_+ = 16291944$) as a regression oracle. Runtime 2.5 s.

4. `codes/vacuum/dr2_cross_energy_lemma.py` (6/6) — the bilinear cross-energy lemma $E_+(A, B) \leq 3|A||B|$ and the off-diagonal pencil bound.
5. `codes/vacuum/dr2_weighted_energy.py` (3/3) — the weighted Lemma A ($\sum_t w_t^2 \leq (1 + T')\|c\|_2^4$).
6. `codes/vacuum/dr2_decoupling_iteration.py` (4/4) — decoupling-route cross-check (NOT load-bearing).
7. `codes/vacuum/dr2_decoupling_exponent.py` (4/4) — decoupling-route exponent cross-check (NOT load-bearing).
8. `codes/vacuum/dr2_lemma_nt_exhaustive.py` (5/5; NEW v1.4) — the Lemma NT applied chain $r_Q(m) \leq 6d(4R - |m|^2)$ verified EXHAUSTIVELY for all 2,644,976 distinct sums across the six shells, ramified corner included; 1.2 s.

Supporting / integration-layer (NOT an entry script): `codes/vacuum/dr2_hadmcoh_margin.py` — the H-ADM-COH discharge-decision margin (long-running; belongs to the `dr2-hadmcoh-discharge-decision` integration note, run artefact `runs/260608-dr2-hadmcoh-margin/`); not load-bearing for the lattice-class theorem. Manifest correction note: the v1.0 “7 scripts” was an auto-discovered list that dropped `dr2_lattice_divisor.py` (a single-line regex artefact: the headline note lists it on the second line of its reproduction command) and included `dr2_hadmcoh_margin.py`; the present manifest is the corrected, load-bearing-plus-declared-cross-check set. Run the entry manifest from the bundle and diff against `expected/`.

5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) *Attack the scope.* The result is confined to the scope stated in Sec. 1; a referee may push that boundary, which is fair — the package makes no claim outside it. In particular: lattice shells only; arbitrary- Q OPEN; decoupling route T4 conditional cross-check; no full C_{full} claim.
- (β) *Attack the tier / over-claim.* The sharpest objection is the $6d(D)$ representation bound: “Dirichlet class-number formula” alone is not a citation. ADDRESSED in v1.1 (operator-directed): Lemma NT is stated precisely with its provenance, the load-bearing chain uses the relaxed constant C_Λ , and the T7 label is restricted (footer): the bundle is PUBLISHED-CONFIRMED as T7-CANDIDATE MOD LEMMA NT; without the Lemma NT pin it is not a confirmed T7 publication. The claim-ledger tier is the arbiter.
- (γ) *Attack reproduction / self-containment.* Each entry script is self-contained (no runtime run-artefact reads — audited after the SC-SCOPE joint_pairing lesson), carries self-test asserts, and ships in the bundle with its transitive dependencies; a referee runs them and diffs against `expected/`.

Quantitative sanity check (mandatory): the Lemma-A chain is re-derivable from its definitions ($\sum_m r_Q(m) = N^2$ exactly; $r_Q(0) \leq N$); the substitution identity $|2a - m|^2 = 4R - |m|^2$ is checked exactly on every tested configuration; the worst machine ratio $T'/6d = 0.25$ leaves a $\times 4$ headroom below the Lemma NT ceiling; and the falsifier (footer) is concrete and pre-registered (a lattice shell violating the divisor bound, or a counterexample to the rank-2 representation count).

6. Result footer

Result ID:	B5-BEYOND-LAYER-BOUND / DR-2 lattice-class referee package
Precise statement:	For $Q = \text{Lambda} \cap \{ x ^2 = R\}$ (fixed crystallographic lattice), with $N = Q $, $E_+(Q) = \#\{a+b=c+d\}$, $r_Q(m) = \#\{(a,b): a+b=m\}$, $T'(Q) = \max_{m \neq 0} r_Q(m)$: $E_+(Q) \leq (1+T'(Q))N^2$ (Lemma A), $T'(Q) \leq C_{\text{Lambda}} \max_{m \neq 0} d(4R - m ^2)$ (y=2a-m reduction + Lemma NT; $C_{\text{Lambda}} \leq 6$ once the textbook lemma is imported), hence $E_+(Q) \leq_{\text{eps}} R^{\text{eps}} N^2$, decoupling-free; Gauss-typical $R \sim N^2$ gives $E_+ \leq_{\text{eps}} N^{\{2+\text{eps}\}}$ (eps redefined, $R^{\text{eps}} \sim N^{\{2\text{eps}\}}$).
Scope:	lattice shells $Q = \text{Lambda} \cap \{ x ^2 = R\}$ ONLY. Arbitrary non-lattice Q OPEN; decoupling route T4 conditional (cross-check, not load-bearing); not a full C_{full} closure.
Evidence grade:	T7-CANDIDATE on the lattice class; Lemma NT PINNED IN-BUNDLE (unramified: full proof, $24 d(D)^2$; ramified corner: cited + outlined + exhaustively machine-covered on the applied domain). EXECUTED (8 entry scripts, 38/38 asserts; lemma_nt_exhaustive NEW, 2,644,976 sums, worst 0.250).
Reproduction command:	python codes/vacuum/dr2_circle_richness.py ; codes/vacuum/dr2_divcirc_proof.py ; codes/vacuum/dr2_lattice_divisor.py ; codes/vacuum/dr2_cross_energy_lemma.py ; codes/vacuum/dr2_weighted_energy.py ; codes/vacuum/dr2_decoupling_iteration.py ; codes/vacuum/dr2_decoupling_exponent.py ; codes/vacuum/dr2_lemma_nt_exhaustive.py
Falsifier:	a lattice shell with E_+ exceeding the divisor bound; or a counterexample to the rank-2 representation count $r \leq C_{\text{NT}} d(D)$.
Tier before / after:	PUBLISHED-BUNDLE CONFIRMED AS T7-CANDIDATE MOD LEMMA NT (operator CONFIRM 2026-06-12); bundle DR2-Lattice-T7Candidate-260612; clean-run confirmed (7/7, 33/33). Final T7 promotion HELD on the Lemma NT pin ONLY. No B5 claim-tier promotion.
No-overclaim statement:	lattice class only, modulo the explicitly stated Lemma NT (constant relaxed to C_{Lambda} in the load-bearing chain); arbitrary non-lattice Q is OPEN; the decoupling route is a non-load-bearing cross-check.
Next required action:	operator review of the Lemma NT pin note (dr2-lemma-nt-inbundle v1.0) + one-script re-run (dr2_lemma_nt_exhaustive.py, 1.2 s) -> final T7 promotion decision + bundle re-stamp. Then per policy sec.15: T-013 SYNTHESIS capstone.